

Biomedical Entity Extraction and Explanation from Patient Reports using BioBERT and MedQuAD

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Abstract—This project presents an end-to-end biomedical text processing system that extracts clinically relevant entities from patient reports using a fine-tuned BioBERT model. The system identifies disease and chemical entities from unstructured medical text and improves interpretability by retrieving related explanations from the MedQuAD dataset. Synthetic patient reports generated using Synthea are used as test inputs for demonstrating the full pipeline. A Gradio-based interface enables real-time user interaction through file upload or direct text entry. Experimental results show strong performance in biomedical named entity recognition, with the system achieving high recall and strong overall F1-score on the BC5CDR benchmark. The resulting prototype demonstrates a complete workflow from raw input to structured and interpretable output.

I. PROJECT SUMMARY

The objective of this project is to design and implement an end-to-end biomedical natural language processing system capable of extracting clinically relevant information from unstructured patient reports. In real medical settings, a large portion of useful information is recorded in free-text form. Although such text contains important details about diseases, medications, and observations, it is difficult to directly use for downstream analysis unless it is converted into structured information. This project addresses that challenge by building a pipeline that identifies important biomedical entities and augments them with explanatory information.

Since Deliverable 1, the project has progressed from early dataset exploration and design planning into a functional prototype. A BioBERT-based Named Entity Recognition (NER) model has been successfully fine-tuned on the BC5CDR dataset for biomedical entity extraction. The system is now capable of accepting raw text input, detecting disease and chemical entities, grouping them into interpretable categories, and retrieving related medical explanations from the MedQuAD dataset. A Gradio-based interface has also been developed so that users can interact with the system in real time.

The implementation now supports the complete path from input to output: a user uploads a patient report or enters medical text, the text is preprocessed, the fine-tuned BioBERT model performs entity extraction, the retrieved entities are categorized, MedQuAD is queried for additional contextual explanation, and the final results are shown in a structured interface. This marks a substantial step forward from Deliverable 1, where the focus was primarily on data acquisition, planning, and initial modeling direction.

At the current stage, the system already demonstrates the essential characteristics of a working machine learning prototype: it can train, evaluate, generate predictions, and present results through an interface. However, some important improvements remain open for the next phase. The current explanation retrieval strategy is based on lexical matching rather than semantic similarity, meaning that the retrieved MedQuAD explanations are useful but not always perfectly context-aware. In addition, although the NER model performs well overall, there is still room to improve boundary detection and reduce false positives, particularly for disease spans that are more context-dependent than chemical names.

Overall, this deliverable demonstrates that the project foundation has successfully been transformed into a functional biomedical NLP system with measurable performance and an accessible interface.

II. SYSTEM ARCHITECTURE AND PIPELINE

The system is designed as a structured, end-to-end biomedical text processing pipeline that transforms unstructured patient reports into meaningful and interpretable outputs. The architecture integrates multiple components, including data generation, preprocessing, machine learning, and knowledge retrieval, to ensure both functionality and interpretability.

The pipeline begins with two primary input sources: user-provided data and synthetically generated patient reports. Users interact with the system through a Gradio-based interface, where they can either upload a patient report in text format or manually enter biomedical text. In parallel, synthetic patient reports generated using Synthea serve as an additional data source for testing and validating the system. These reports simulate real-world clinical data and help evaluate the robustness of the pipeline under realistic conditions.

Both user input and Synthea-generated reports are directed to the input module, which standardizes the data format and prepares it for further processing. The text then undergoes preprocessing, where it is tokenized and formatted to match the requirements of the BioBERT model. This step is critical because transformer-based models rely on subword tokenization, and proper alignment between tokens and labels is necessary to maintain the integrity of entity boundaries.

The core component of the system is the BioBERT-based Named Entity Recognition (NER) model, which has been fine-tuned on the BC5CDR dataset. BioBERT is specifically designed for biomedical text and leverages domain-specific

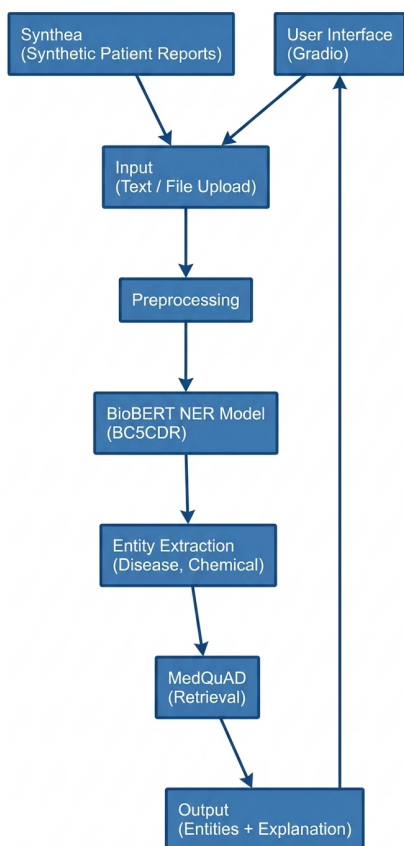


Fig. 1. System architecture showing the full pipeline from user input to structured output with entity extraction and explanation retrieval.

pretraining to improve performance on medical language tasks. During inference, the model performs token-level classification using the BIO tagging scheme to identify entities corresponding to diseases and chemical substances.

Following model inference, an entity extraction module processes the token-level predictions to reconstruct complete entity phrases. This step merges subword tokens and groups identified entities into meaningful categories such as diseases and chemicals. The result is a structured representation of the key biomedical information present in the input text.

To enhance interpretability, the system integrates a knowledge retrieval component using the MedQuAD dataset. For each extracted entity, the system attempts to retrieve a relevant question-answer pair from MedQuAD. This provides contextual explanations that help users understand the significance of the extracted entities. Although the current implementation relies on keyword-based matching, it establishes a clear connection between entity recognition and external biomedical knowledge.

The final stage of the pipeline is the output module, which presents the results through the user interface. The output includes a detailed list of extracted entities, grouped summaries by category, and explanatory information retrieved from MedQuAD. A concise summary is also generated to

provide a quick overview of the findings.

An important aspect of the architecture is the feedback loop between the output and the user interface. Once results are displayed, users can refine their input or test new reports, enabling an interactive and iterative workflow.

As illustrated in Fig. 1, the pipeline follows a clear and linear flow from input to output, while incorporating multiple data sources and processing stages. The inclusion of Synthesia as a synthetic data generator highlights the system’s ability to operate on realistic test data, while the integration of MedQuAD demonstrates the added value of combining entity extraction with knowledge retrieval.

This architecture is modular and extensible, allowing individual components such as the retrieval mechanism or preprocessing pipeline to be improved independently without disrupting the overall system.

III. SYSTEM REFINEMENTS AND IMPROVEMENTS

This phase of the project focuses on improving the usability, stability, and overall robustness of the biomedical text processing system. While the core pipeline remains unchanged, several enhancements have been introduced to strengthen the system as a complete and reliable application.

One major improvement involves the model deployment process. Instead of repeatedly initializing or retraining the model during runtime, the fine-tuned BioBERT model is now stored persistently and loaded directly from disk. This significantly reduces execution time, improves consistency across runs, and ensures that the system behaves reliably in different sessions.

The MedQuAD retrieval component has also been refined. Earlier implementations relied on simple keyword matching, which occasionally produced inconsistent results due to variations in dataset structure. The retrieval logic has now been extended to consider multiple fields such as question focus and synonyms, resulting in more stable and contextually relevant explanations for extracted entities.

The user interface has undergone a substantial redesign. The earlier version provided basic functionality for input and output display. The updated interface adopts a more structured layout using clearly defined sections for input, detailed results, grouped outputs, and summaries. Additional interaction controls, including dedicated “Analyze” and “Clear” buttons, improve usability and make the system more intuitive to operate.

Output presentation has been enhanced to improve interpretability. Extracted entities are now displayed in a cleaner tabular format with associated explanations, followed by grouped summaries that organize entities into meaningful biomedical categories. A concise summary section provides an overview of detected entities, improving accessibility for non-technical users.

System stability has also been strengthened by ensuring that all required components, including the model, tokenizer, and datasets, are properly loaded and reused across sessions. This

reduces runtime errors and improves the overall reliability of the pipeline.

IV. MODEL IMPLEMENTATION DETAILS

The implementation is built primarily with the Hugging Face Transformers framework and PyTorch. BioBERT, specifically `dmis-lab/biobert-base-cased-v1.2`, is used as the underlying language model. This choice is motivated by the fact that BioBERT is pre-trained on biomedical literature, making it better aligned with medical terminology than a general-purpose BERT model.

The NER training stage uses the BC5CDR dataset, which contains annotations for chemical and disease entities. Since the dataset is used for token classification, the data must be converted into a format compatible with transformer-based training. This involves constructing token sequences, aligning word-level labels to subword tokens, and creating the final training examples used by the model.

One of the most important technical details is **token-label alignment**. Because the tokenizer may split a biomedical term into multiple subword tokens, the implementation ensures that labels remain consistent across the resulting token sequence. For continuation subwords, the implementation adjusts BIO labels appropriately so that the model learns consistent entity boundaries. This step is essential for preserving entity structure during both training and inference.

The training configuration used in this deliverable is summarized below:

- **Model:** BioBERT (`dmis-lab/biobert-base-cased-v1.2`)
- **Task:** Token classification for biomedical named entity recognition
- **Training dataset:** BC5CDR
- **Epochs:** 3
- **Learning rate:** 2×10^{-5}
- **Batch size:** 8
- **Optimizer:** AdamW through the Hugging Face Trainer pipeline

The model is trained using the Hugging Face Trainer API, which simplifies batching, evaluation, logging, and checkpoint management. During training, the implementation evaluates the model on the validation split and selects the best-performing checkpoint based on F1-score. This improves training reliability and ensures that the saved model is useful for downstream inference.

To support reproducibility, the project includes modular code for the following tasks:

- dataset loading and preprocessing
- tokenization and label alignment
- model training and checkpoint saving
- evaluation metrics generation
- confusion matrix and classification report creation
- inference over uploaded patient reports
- MedQuAD retrieval and Gradio interface integration

This modular design is important because it separates the concerns of data preparation, model learning, evaluation, and

deployment. As a result, future work can replace or improve one component without requiring a full redesign of the pipeline.

V. INTERFACE PROTOTYPE

An interactive interface was developed using Gradio to make the model accessible for demonstration and testing. The decision to use Gradio was motivated by its simplicity, ease of deployment, and strong support for rapid prototyping of machine learning applications.

The interface supports two input modes:

- 1) uploading a patient report in `.txt` format
- 2) directly entering biomedical text into a text box

This flexibility is valuable because it allows the system to support both realistic report-based workflows and quick testing with custom input text. Once the user submits input, the interface triggers the full inference pipeline: text processing, entity prediction using BioBERT, grouping of entities, explanation retrieval using MedQuAD, and summary generation.

The interface provides four main output views:

- **Input Text:** shows the full report or entered text
- **Extracted Entities + MedQuAD Explanations:** a detailed table listing each entity, its type, and the matched MedQuAD question
- **Grouped Output:** summarizes entities by category, such as diseases and chemicals
- **Summary:** reports total entities found and lists extracted categories in a concise format

This interface design improves usability because it presents information in layers. A technical user can inspect the detailed table, while a non-technical user can focus on the grouped output and summary. This layered presentation makes the system easier to interpret and more effective as a demonstration tool.

Fig. 2 shows the user interface with file upload and text entry capability. Fig. 3 shows the output after processing a patient report.

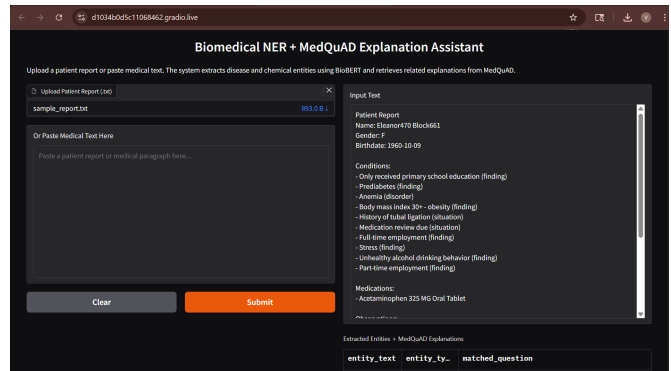


Fig. 2. Gradio interface showing report upload and text input for testing the model.

The interface also reveals some current limitations. The extracted MedQuAD explanations are informative, but not always perfectly aligned with the exact context of the report.

Extracted Entities + MedQuAD Explanations

entity_text	entity_ty	matched_question
Prediabetes	Disease	What is (are) Prediabetes ?
Anemia	Disease	What is (are) dyserythropoietic anemia and thrombocytopenia
obesity	Disease	What are the symptoms of Intellectual disability, epileptic
alcohol	Chemical	What is (are) Acute alcohol sensitivity ?
Acetaminophen	Chemical	What is (are) Acetaminophen and codeine overdose ?
Pain	Disease	What is (are) paroxysmal extreme pain disorder ?
Glucose	Chemical	What is (are) glucose phosphate isomerase deficiency ?

Grouped Output

Category	Entities
Diseases	Prediabetes, Anemia, obesity, Pain
Chemicals	alcohol, Acetaminophen, Glucose

Summary

Total entities found: 7
Diseases found: 4
Chemicals found: 3
Diseases: Prediabetes, Anemia, obesity, Pain
Chemicals: alcohol, Acetaminophen, Glucose

Flag

Fig. 3. Interface output showing extracted entities, grouped categories, and summary information.

This is because the present retrieval strategy depends on lexical similarity rather than semantic relevance. In future iterations, a semantic retrieval module could substantially improve explanation quality.

VI. EARLY EVALUATION AND RESULTS

The model was evaluated on the BC5CDR test dataset using standard metrics for sequence labeling tasks. The primary evaluation results are:

- **Precision:** 0.8416
- **Recall:** 0.8919
- **F1-score:** 0.8660
- **Accuracy:** 0.9665

Fig. 4 shows the evaluation metric summary.

These results indicate that the model performs strongly on biomedical entity recognition. The most important value here is the F1-score of 0.8660, because it balances precision and recall and is more informative than raw accuracy for imbalanced token classification tasks.

The recall value of 0.8919 is especially significant. It indicates that the model is able to capture a large proportion of true biomedical entities. In a biomedical text processing context, this is a valuable property because missing an important disease or chemical mention can reduce the usefulness of the system. High recall therefore suggests that the model is effective at discovering relevant biomedical information from the input text.

The precision value of 0.8416 is slightly lower than recall, which suggests that the model occasionally predicts entities

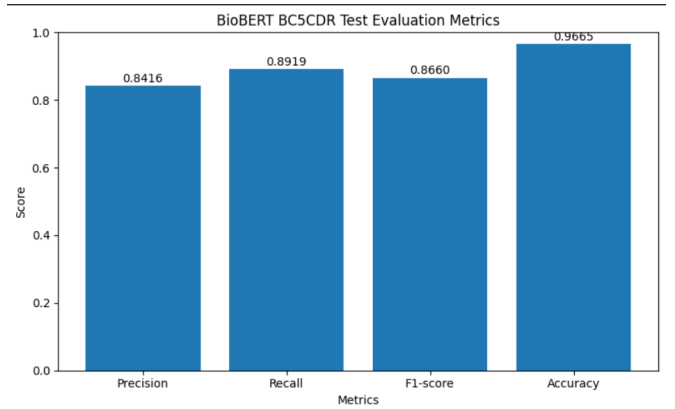


Fig. 4. Evaluation metrics of the fine-tuned BioBERT model on the BC5CDR test set.

where none are present. In practical terms, this means the system is somewhat more willing to highlight a token span as an entity. This is often a reasonable trade-off for early-stage information extraction systems, especially when recall is important. However, it also indicates that future work should focus on reducing false positives, particularly for disease-related spans.

Although the overall accuracy is very high at 0.9665, accuracy must be interpreted carefully in token classification. Most tokens in biomedical text belong to the $\emptyset$ class, meaning that they are not part of an entity. As a result, a model can achieve a high token-level accuracy even if entity-level performance is not perfect. Therefore, precision, recall, and F1-score are more informative measures of actual system quality.

The classification report gives more detailed insight into class-wise behavior. Chemical entities show particularly strong performance, with high precision and recall for both beginning and inside tags. Disease entities also perform well, but slightly lower than chemical entities. This difference is reasonable because disease mentions often have more varied linguistic structure and may depend more strongly on surrounding context. Chemical names, by contrast, often have stronger lexical regularities that make them easier for the model to detect.

Fig. 5 presents the confusion matrix.

The confusion matrix shows that the vast majority of predictions lie on the diagonal, which indicates correct classification across all major labels. This is strong evidence that the model has learned the structure of the task effectively. The most common remaining errors occur near entity boundaries, especially between beginning and inside labels. This is a typical challenge in sequence labeling because the model must not only recognize the existence of an entity, but also determine exactly where it starts and ends.

In addition to quantitative evaluation, qualitative results from the interface are also encouraging. When tested on a synthetic patient report, the system correctly extracted disease entities such as *Prediabetes*, *Anemia*, and *obesity*, as well as chemical entities such as *Acetaminophen*, *Glucose*, and *alcohol*. The grouped output and summary make these results

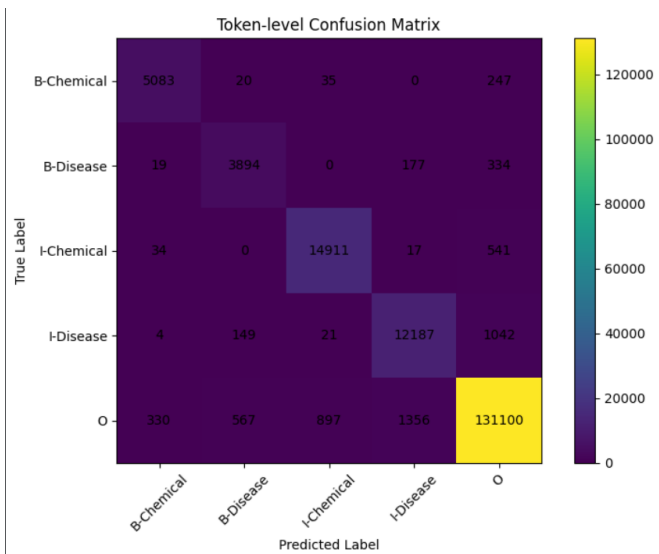


Fig. 5. Token-level confusion matrix showing classification behavior across biomedical entity labels.

easier to interpret, while the MedQuAD retrieval component provides additional context.

These qualitative outputs show that the prototype is not limited to benchmark evaluation alone. It is capable of processing realistic report-style input and returning structured biomedical information in a usable form. This is particularly important for demonstrating that the project functions end-to-end rather than only as a standalone model training exercise.

Overall, the results suggest that the model is already effective enough to support a working biomedical entity extraction prototype. The strongest evidence comes from the combination of benchmark-level performance, low off-diagonal confusion in the matrix, and successful qualitative outputs through the interface.

VII. INTERFACE COMPARISON AND USABILITY IMPROVEMENTS

To improve the practical usability of the system, the user interface was refined to provide a clearer layout, better interaction flow, and more interpretable outputs. While the earlier version of the interface was functional, it primarily focused on basic input-output behavior without strong structural organization.

The updated interface introduces a more organized layout with clearly separated sections for input text, extracted entities, grouped biomedical categories, and summary information. This structured design allows users to better understand the output at different levels of detail, making the system more accessible to both technical and non-technical users.

Fig. 6 shows the refined interface and output for a sample patient report.

In addition to layout improvements, interaction controls were enhanced by introducing dedicated “Analyze” and “Clear” buttons. These controls provide a more intuitive workflow by allowing users to explicitly trigger processing and reset

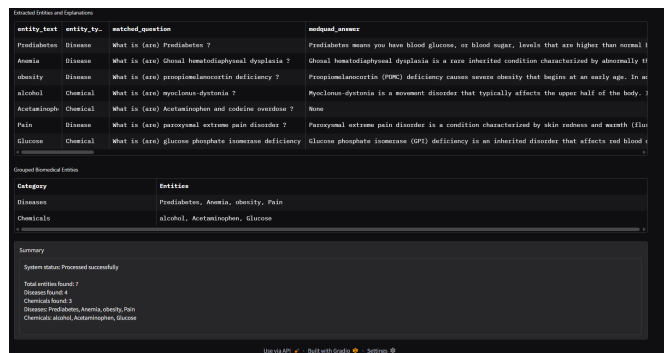


Fig. 6. Refined interface showing improved layout, grouped outputs, and summary generation.

inputs when needed. This results in a more predictable and user-friendly interaction compared to the earlier single-step submission flow.

The output presentation has also been significantly improved. Extracted entities are now displayed in a structured tabular format along with associated explanations retrieved from MedQuAD. Furthermore, the system groups entities into meaningful biomedical categories such as diseases and chemicals, improving interpretability. A summary section provides a concise overview of detected entities, including counts and categorized listings.

Table I summarizes the key usability improvements introduced in the refined interface.

TABLE I
INTERFACE USABILITY IMPROVEMENTS

Aspect	Earlier Interface	Refined Interface
Layout	Basic layout	Structured multi-section layout
User Controls	Single submission flow	Analyze and Clear buttons
Output Display	Flat entity list	Detailed + grouped outputs
Interpretability	Limited output clarity	Summary + categorized entities
Usability	Functional	More intuitive and user-friendly

Overall, these improvements enhance the usability and interpretability of the system, making it more effective as a practical biomedical text processing tool. The refined interface supports clearer interaction, better output understanding, and improved user experience without altering the underlying model.

VIII. CHALLENGES AND NEXT STEPS

Several practical and technical challenges were encountered during the implementation of this system.

One major challenge was the integration of multiple datasets with distinct roles and formats. BC5CDR is used for supervised NER training, Synthea is used to generate realistic synthetic patient reports, and MedQuAD is used as a retrieval-based explanation resource. Because these datasets differ substantially in structure and purpose, integrating them into a single coherent workflow required careful planning and repeated debugging.

Another challenge came from the development environment. The system was prototyped in Google Colab, where runtime resets can erase intermediate files and variables. This led to repeated regeneration of data and reloading of models, which complicated the workflow. As a result, saving intermediate datasets, trained models, and results to Google Drive became an important part of the implementation process.

A third challenge involves the explanation module. While the MedQuAD integration improves interpretability, the current retrieval approach is based on keyword overlap. This sometimes produces explanations that are only loosely related to the entity as it appears in the report. The system therefore demonstrates the usefulness of retrieval-based explanation, but also highlights the need for more advanced retrieval methods.

The next stage of the project will focus on several improvements:

- replacing keyword-based retrieval with embedding-based semantic search
- improving precision by tuning hyperparameters and refining post-processing
- improving boundary detection for multi-token disease mentions
- polishing the user interface for clearer display and better user experience
- expanding the system toward question answering or more advanced biomedical reasoning

These next steps are intended to move the system from a strong prototype toward a more robust and context-aware biomedical assistant.

IX. RESPONSIBLE AI REFLECTION

This project operates in the biomedical domain, where accuracy, interpretability, and responsible use are especially important. Although the system can identify disease and chemical entities from patient reports, it is not designed to provide diagnoses, treatment plans, or medical advice. Its outputs should therefore be understood as informational rather than clinical.

One ethical concern involves **misinterpretation**. Because the system can extract medically relevant terms and attach explanations, users may incorrectly assume that it is performing diagnosis or clinical reasoning. For this reason, it is important to clearly communicate that the system is an entity extraction and explanation prototype rather than a decision-making medical tool.

Another concern is **dataset bias**. The BC5CDR dataset, while standard and useful, may not represent the full diversity of biomedical language, patient populations, or medical conditions. As a result, the model may perform better on entity types and phrasing patterns that are well represented in the training data than on those that are not.

A third issue concerns the **retrieval quality of explanations**. Since the current MedQuAD integration uses keyword matching, it may return explanations that are partially relevant but not fully aligned with the report context. In a biomedical

setting, such mismatches could be misleading if presented without sufficient caution.

To address these issues, future work will emphasize better semantic retrieval, clearer uncertainty communication, and stronger statements about system limitations in the interface and documentation. Responsible deployment requires not only technical improvement, but also careful communication of what the system can and cannot do.

X. CONCLUSION

This project successfully demonstrates a complete biomedical NLP pipeline that integrates entity extraction and explanation retrieval. By fine-tuning BioBERT on BC5CDR and combining it with MedQuAD-based retrieval, the system is able to process unstructured patient reports and return structured biomedical information in an interpretable form.

The addition of a Gradio interface makes the system interactive and practical for demonstration. Quantitative results show that the model performs strongly on the benchmark task, while qualitative results demonstrate that the full pipeline works on realistic synthetic reports. Together, these results confirm that the project has moved beyond dataset exploration into a functioning prototype.

Although there are still important improvements to be made, especially in semantic retrieval and precision refinement, the current implementation already provides a strong foundation for future biomedical AI applications. The system demonstrates how domain-specific language modeling, structured output generation, and knowledge retrieval can be combined to produce a useful and interpretable biomedical text processing tool.